

Random variables

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Random variables

Definition 0.1 (Random variable). A **random variable** is a function from a [sample space](#) Ω to the real numbers $\mathbb{R}$: it assigns a number to each possible outcome of a random experiment.

We use uppercase letters ($X, Y, Z, \dots$) for random variables, and lowercase letters ($x, y, z, \dots$) for their realized (observed) values. We write $\{X = x\}$ for the event $\{\omega \in \Omega : X(\omega) = x\}$, and similarly for $\{X \leq x\}$ and other conditions on X . A fully rigorous definition also requires the function to be *measurable*, so that every such set is an event with a probability; that requirement never binds in these notes.

See also: https://en.wikipedia.org/wiki/Random_variable

Example 0.1 (Number of heads in two coin flips). Flip a fair coin twice. The sample space is $\Omega = \{HH, HT, TH, TT\}$, each outcome with probability $1/4$. Let X be the number of heads:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0$$

Then $\{X = 1\} = \{HT, TH\}$, so $\Pr(X = 1) = 1/4 + 1/4 = 1/2$.

Definition 0.2 (Range of a random variable). The **range** of a random variable X , denoted $\mathcal{R}(X)$, is the set of values X can take:

$$\mathcal{R}(X) \stackrel{\text{def}}{=} \{X(\omega) : \omega \in \Omega\}$$

Example 0.2 (Range of the number of heads). In Example [0.1](#), $\mathcal{R}(X) = \{0, 1, 2\}$.

Definition 0.3 (Discrete random variable). A random variable X is **discrete** if its [range](#) $\mathcal{R}(X)$ is finite or countably infinite.

Example 0.3 (Discrete random variables). The number of heads in [Example 0.1](#) is discrete, with a finite range. A count with no upper bound, such as the number of emergency-department visits in a day, is discrete with a countably infinite range, $\{0, 1, 2, \dots\}$.

Definition 0.4 (Continuous random variable). A random variable X is **continuous** if $\Pr(X = x) = 0$ for every $x \in \mathbb{R}$.

Every continuous random variable in these notes also has a [probability density function](#). Continuous random variables without a density exist, but they do not arise in applications. Some random variables are neither discrete nor continuous: a time to event that equals exactly 0 with positive probability, and otherwise spreads over $(0, \infty)$, is one.

Definition 0.5 (Uniform distribution). A random variable X has the **uniform distribution** on an interval $[\alpha, \beta]$, with $\alpha < \beta$, written $X \sim \text{Uniform}(\alpha, \beta)$, if the probability that X falls in any subinterval of $[\alpha, \beta]$ is proportional to the length of that subinterval:

$$\Pr(a \leq X \leq b) = \frac{b - a}{\beta - \alpha} \quad \text{for all } \alpha \leq a \leq b \leq \beta$$

Example 0.4 (A uniformly distributed random variable). Let $X \sim \text{Uniform}(0, 1)$ ([Definition 0.5](#)), so $\Pr(a \leq X \leq b) = b - a$ for all $0 \leq a \leq b \leq 1$. Taking $a = b = x$ gives $\Pr(X = x) = x - x = 0$ for every $x \in [0, 1]$. Taking $a = 0$ and $b = 1$ gives $\Pr(0 \leq X \leq 1) = 1$, so by the [complement rule](#) X falls outside $[0, 1]$ with probability 0, and $\Pr(X = x) = 0$ for every x outside $[0, 1]$ as well. So X is continuous. Its range, $[0, 1]$, is uncountable.

Characteristics of probability distributions

Definition 0.6 (Probability mass function (PMF)). If X is a [discrete random variable](#), the **probability mass function** of X at value x , denoted $f(x)$, $f_X(x)$, $P(x)$, $P_X(x)$, or $P(X = x)$, is the probability that X takes exactly the value x :

$$P(X = x) \stackrel{\text{def}}{=} \Pr(\{X = x\})$$

See also https://en.wikipedia.org/wiki/Probability_mass_function

Example 0.5 (PMF of the number of heads). For the number of heads X in two fair coin flips (Example 0.1):

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}$$

Definition 0.7 (Bernoulli distribution). A discrete random variable X has the **Bernoulli distribution** with parameter $\pi \in [0, 1]$, written $X \sim \text{Ber}(\pi)$, if:

$$\begin{aligned} \Pr(X = x) &\stackrel{\text{def}}{=} 1_{x \in \{0,1\}} \pi^x (1 - \pi)^{1-x} \\ &= \begin{cases} \pi, & x = 1 \\ 1 - \pi, & x = 0 \end{cases} \end{aligned}$$

Example 0.6 (The first of two coin flips). In Example 0.1, let X_1 indicate heads on the first flip, so $X_1(HH) = X_1(HT) = 1$ and $X_1(TH) = X_1(TT) = 0$. Then $P(X_1 = 1) = \Pr(\{HH, HT\}) = 1/2$ and $P(X_1 = 0) = \Pr(\{TH, TT\}) = 1/2$, so $X_1 \sim \text{Ber}(1/2)$.

Definition 0.8 (Probability density function (PDF)). If X is a continuous random variable, a **probability density function** of X , denoted $f(x)$, $f_X(x)$, $p(x)$, $p_X(x)$, or $p(X = x)$, is a function $f \geq 0$ whose integral over any interval is the probability that X falls in that interval:

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx \quad \text{for all } a \leq b$$

A density is not a probability: it can exceed 1. A density is also not unique: changing f at finitely many points changes none of its integrals, so the value of a density at a single point carries no probability by itself. These notes use the version that is continuous wherever possible. Theorem 0.3 gives the value of f at a point where it is continuous.

See also https://en.wikipedia.org/wiki/Probability_density_function#Formal_definition

Example 0.7 (Density of a uniform random variable). Let $X \sim \text{Uniform}(0, 1)$ (Example 0.4), and let $f(x) = 1$ for $x \in [0, 1]$ and $f(x) = 0$ otherwise. For $0 \leq a \leq b \leq 1$:

$$\begin{aligned} \int_a^b f(x) dx &= \int_a^b 1 dx && (f = 1 \text{ on } [0, 1]) \\ &= b - a && (\text{integrate}) \\ &= \Pr(a \leq X \leq b) && (\text{definition of the uniform distribution}) \end{aligned}$$

An interval reaching outside $[0, 1]$ adds nothing to either side: $f = 0$ there, and X falls outside $[0, 1]$ with probability 0 (Example 0.4). So f is a density of X .

Definition 0.9 (Normal distribution). A random variable X has the **normal distribution** (or **Gaussian distribution**) with mean parameter $\mu \in \mathbb{R}$ and variance parameter $\sigma^2 > 0$, written $X \sim N(\mu, \sigma^2)$, if X is continuous with **density**:

$$p(X = x) \stackrel{\text{def}}{=} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$$

The distribution $N(0, 1)$ is called the **standard normal distribution**. This density integrates to 1, and a normal random variable has mean μ and variance σ^2 , which is where the parameters' names come from (Casella and Berger 2002).

Example 0.8 (Normal densities at their centers). For $X \sim N(0, 1)$, the density at $x = 0$ is:

$$\begin{aligned} p(X = 0) &= \frac{1}{1 \cdot \sqrt{2\pi}} e^{-\frac{(0-0)^2}{2 \cdot 1}} && \text{(normal density with } \mu = 0, \sigma^2 = 1) \\ &= \frac{1}{\sqrt{2\pi}} && (e^0 = 1) \\ &\approx 0.399 && \text{(evaluate)} \end{aligned}$$

For $X \sim N(0, 0.1^2)$, the same steps give $p(X = 0) = 1/(0.1\sqrt{2\pi}) \approx 3.99$, a density greater than 1.

Definition 0.10 (Exponential distribution). A random variable T has the **exponential distribution** with rate $\lambda > 0$ if T is continuous with **density**:

$$f(t) \stackrel{\text{def}}{=} \begin{cases} \lambda e^{-\lambda t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Example 0.9 (Probability of an event within one time unit). Let T be exponential with rate λ . The probability that T falls in $[0, 1]$ is:

$$\begin{aligned} \Pr(0 \leq T \leq 1) &= \int_0^1 \lambda e^{-\lambda t} dt && \text{(definition of a density)} \\ &= [-e^{-\lambda t}]_{t=0}^1 && \text{(antiderivative of } \lambda e^{-\lambda t}) \\ &= 1 - e^{-\lambda} && \text{(evaluate at the bounds)} \end{aligned}$$

With $\lambda = 0.5$, for example, $\Pr(0 \leq T \leq 1) = 1 - e^{-0.5} \approx 0.393$.

Definition 0.11 (Cumulative distribution function (CDF)). For a random variable X (discrete or continuous), its **cumulative distribution function** is:

$$F(t) \stackrel{\text{def}}{=} \Pr(X \leq t), \quad t \in \mathbb{R}.$$

The CDF is always defined, whether X is discrete or continuous, and it fully characterizes X 's distribution. It is non-decreasing, with $F(t) \rightarrow 0$ as $t \rightarrow -\infty$ and $F(t) \rightarrow 1$ as $t \rightarrow \infty$.

See also https://en.wikipedia.org/wiki/Cumulative_distribution_function

Example 0.10 (CDF of the number of heads). Summing the PMF in Example 0.5 over values at or below t :

$$F(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{4}, & 0 \leq t < 1 \\ \frac{3}{4}, & 1 \leq t < 2 \\ 1, & t \geq 2 \end{cases}$$

Definition 0.12 (Quantile function (population inverse CDF)). For a random variable X with [cumulative distribution function \(CDF\)](#) F , its population **quantile function** (generalized inverse of F) is:

$$Q(p) \stackrel{\text{def}}{=} \inf\{t : F(t) \geq p\}, \quad 0 < p < 1.$$

The restriction to $0 < p < 1$ keeps $Q(p)$ finite: for $p = 1$, the set $\{t : F(t) \geq 1\}$ is empty whenever $F(t) < 1$ for every t , as for a [normal distribution](#), and the infimum of the empty set is $+\infty$.

Example 0.11 (Median of the number of heads). From Example 0.10, $F(t) \geq 0.5$ first holds at $t = 1$, so the median of the number of heads is $Q(0.5) = 1$.

Theorem 0.1 (A density integrates to the CDF, and is its derivative). *If X is a continuous random variable with density f and CDF F , then for every t :*

$$F(t) = \int_{-\infty}^t f(x) dx$$

and at every t where f is continuous:

$$f(t) = \frac{\partial}{\partial t} F(t)$$

i Proof

Proof. For any $a < t$, $\{X \leq t\}$ is the disjoint union of $\{X \leq a\}$ and $\{a < X \leq t\}$, and $\Pr(X = a) = 0$ because X is continuous. So:

$$\begin{aligned} F(t) &= \Pr(X \leq a) + \Pr(a < X \leq t) && \text{(additivity, and the definition of the CDF)} \\ &= F(a) + \Pr(a \leq X \leq t) && \text{(definition of the CDF; } \Pr(X = a) = 0) \\ &= F(a) + \int_a^t f(x) dx && \text{(definition of the density)} \end{aligned}$$

Letting $a \rightarrow -\infty$, $F(a) \rightarrow 0$ (Definition 0.11), which gives $F(t) = \int_{-\infty}^t f(x) dx$. The fundamental theorem of calculus then gives $\frac{\partial}{\partial t} \int_{-\infty}^t f(x) dx = f(t)$ at every t where f is continuous. $\square$

For the densities in these notes, f is continuous at all but finitely many points, so $F' = f$ at all but finitely many points.

Example 0.12 (CDF and density of a uniform random variable). For X uniform on $[0, 1]$ with the density f of Example 0.7, and $t \in [0, 1]$:

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) dx && \text{(the CDF is the integral of the density)} \\ &= \int_{-\infty}^0 0 dx + \int_0^t 1 dx && \text{(split at 0, and substitute } f) \\ &= t && \text{(integrate)} \end{aligned}$$

For $t \in (0, 1)$, where f is continuous, $\frac{\partial}{\partial t} F(t) = 1 = f(t)$. At $t = 1$, f jumps from 1 to 0, and F has no derivative: its slope is 1 on the left and 0 on the right.

Theorem 0.2 (A piecewise-smooth CDF has its derivative as a density). *If the CDF F of a random variable X is continuous everywhere, and has a continuous derivative at all but finitely many points, then X is continuous, and F' (given any values at the finitely many exceptional points) is a density of X .*

i Proof

Proof. For events $A \subseteq B$, B is the disjoint union of A and $B \setminus A$, so additivity gives $\Pr(A) \leq \Pr(B)$. For any x and $\Delta > 0$, $\{X = x\} \subseteq \{x - \Delta < X \leq x\}$, so:

$$\begin{aligned} 0 &\leq \Pr(X = x) && \text{(probabilities are non-negative)} \\ &\leq \Pr(x - \Delta < X \leq x) && (\{X = x\} \subseteq \{x - \Delta < X \leq x\}) \\ &= F(x) - F(x - \Delta) && \text{(additivity, and the definition of the CDF)} \end{aligned}$$

and $F(x) - F(x - \Delta) \rightarrow 0$ as $\Delta \downarrow 0$, because F is continuous. So $\Pr(X = x) = 0$ for every x , and X is continuous. Then, for $a \leq b$:

$$\begin{aligned} \Pr(a \leq X \leq b) &= \Pr(a < X \leq b) \quad (\Pr(X = a) = 0) \\ &= F(b) - F(a) \quad \text{(additivity, and the definition of the CDF)} \\ &= \int_a^b F'(x) dx \quad \text{(fundamental theorem of calculus)} \end{aligned}$$

The last step applies the fundamental theorem of calculus on each piece of $[a, b]$ between exceptional points, where F' is continuous, and joins the pieces using the continuity of F at the exceptional points. Finally, $F' \geq 0$ wherever it exists, because F is non-decreasing. $\square$

Example 0.13 (A density from a CDF with a corner). Let X have CDF $F(t) = 0$ for $t < 0$, $F(t) = t^2$ for $0 \leq t \leq 1$, and $F(t) = 1$ for $t > 1$. This F is continuous everywhere, and has a continuous derivative everywhere except $t = 1$, where its slope drops from 2 to 0. By Theorem 0.2, X is continuous with density $f(t) = 2t$ for $0 \leq t < 1$ and $f(t) = 0$ otherwise. As a check, $\int_0^1 2t dt = 1$.

Theorem 0.3 (A density as a limit of interval probabilities). *If X is a continuous random variable with density f , and f is continuous at x , then $f(x)$ is the limit of the probability that X falls in an interval starting at x , divided by the width of that interval, as that width shrinks to 0:*

$$f(x) = \lim_{\Delta \downarrow 0} \frac{\Pr(x \leq X < x + \Delta)}{\Delta}$$

i Proof

Proof. For $\Delta > 0$:

$$\begin{aligned}\frac{\Pr(x \leq X < x + \Delta)}{\Delta} &= \frac{\Pr(x < X \leq x + \Delta)}{\Delta} \quad (\Pr(X = x) = \Pr(X = x + \Delta) = 0) \\ &= \frac{F(x + \Delta) - F(x)}{\Delta} \quad (\text{additivity, and the definition of the CDF})\end{aligned}$$

By Theorem 0.1, F is differentiable at x with $F'(x) = f(x)$, so this difference quotient converges to $f(x)$ as $\Delta \downarrow 0$. $\square$

For small $\Delta > 0$, $f(x) \cdot \Delta \approx \Pr(x \leq X < x + \Delta)$. The limit is the right derivative of F at x . At a point where f is not continuous, the limit can differ from the chosen value $f(x)$: for $X \sim \text{Uniform}(0, 1)$, the limit at $x = 1$ is 0, although Example 0.7 sets $f(1) = 1$.

See also Rothman et al. (2021) (Chapter 22, p. 535).

Example 0.14 (The uniform density as a limit). For $X \sim \text{Uniform}(0, 1)$ and $x \in [0, 1]$, $\Pr(x \leq X < x + \Delta) = \Delta$ once $\Delta \leq 1 - x$, so:

$$f(x) = \lim_{\Delta \downarrow 0} \frac{\Delta}{\Delta} = 1$$

For $x < 0$ or $x > 1$, the interval eventually misses $[0, 1]$, so the limit is 0. Both agree with the density of Example 0.7.

Theorem 0.4 (Density functions integrate to 1). *If X is a continuous random variable with density f :*

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Proof

Proof.

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= \lim_{b \rightarrow \infty} \int_{-\infty}^b f(x) dx \quad (\text{definition of an improper integral}) \\ &= \lim_{b \rightarrow \infty} F(b) \quad (\text{the CDF is the integral of the density}) \\ &= 1 \quad (\text{limit of a CDF})\end{aligned}$$

$\square$

Example 0.15 (The uniform density integrates to 1). For X uniform on $[0, 1]$ (Example 0.7), $\int_{-\infty}^{\infty} f(x) dx = \int_0^1 1 dx = 1$.

Definition 0.13 (Jointly distributed random variables). Random variables $X_1, \dots, X_n$ are **jointly distributed** when they are defined on the same **sample space** Ω , so that an event about several of them at once, such as $\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}$, has a probability.

We write $\Pr(X \in A, Y \in B)$ for $\Pr(\{X \in A\} \cap \{Y \in B\})$, and similarly for more variables. The collection of all such probabilities is the **joint distribution** of the variables.

Example 0.16 (The first flip and the total). In Example 0.1, let X_1 indicate heads on the first flip (as in Example 0.6), and let X be the total number of heads. Both are functions on the same sample space, $\{HH, HT, TH, TT\}$, so they are jointly distributed, and, for example:

$$\Pr(X_1 = 1, X = 1) = \Pr(\{HH, HT\} \cap \{HT, TH\}) = \Pr(\{HT\}) = \frac{1}{4}$$

Definition 0.14 (Joint probability mass function). For **jointly distributed** discrete random variables X and Y , the **joint probability mass function** (joint PMF) of X and Y is the probability that X takes the value x and Y takes the value y :

$$P(X = x, Y = y) \stackrel{\text{def}}{=} \Pr(\{X = x\} \cap \{Y = y\})$$

Example 0.17 (Joint PMF of the first flip and the total). Continuing Example 0.16, each outcome of the two flips fixes both X_1 and X , so the joint PMF puts probability $1/4$ on each outcome's pair of values:

	$X = 0$	$X = 1$	$X = 2$
$X_1 = 0$	$1/4$ (TT)	$1/4$ (TH)	0
$X_1 = 1$	0	$1/4$ (HT)	$1/4$ (HH)

Definition 0.15 (Marginal distribution). When a random variable X is one of several **jointly distributed** random variables, the distribution of X by itself (its CDF, and its PMF or density) is called the **marginal distribution** of X .

The word “marginal” only says that the other variables are being set aside; the marginal distribution of X is the same object as the distribution of X .

Example 0.18 (Marginal distribution of the total). In Example 0.17, the marginal distribution of the total X is the PMF of Example 0.5: $P(X = 0) = 1/4$, $P(X = 1) = 1/2$, $P(X = 2) = 1/4$.

Theorem 0.5 (Marginal PMF from a joint PMF). *If X and Y are jointly distributed discrete random variables, then for every x :*

$$P(X = x) = \sum_{y \in \mathcal{R}(Y)} P(X = x, Y = y)$$

i Proof

Proof. The event $\{X = x\}$ is the disjoint union of the events $\{X = x\} \cap \{Y = y\}$ over the countably many values $y \in \mathcal{R}(Y)$, so:

$$\begin{aligned} P(X = x) &= \Pr\left(\bigcup_{y \in \mathcal{R}(Y)} (\{X = x\} \cap \{Y = y\})\right) && \text{(the events partition } \{X = x\}) \\ &= \sum_{y \in \mathcal{R}(Y)} \Pr(\{X = x\} \cap \{Y = y\}) && \text{(countable additivity)} \\ &= \sum_{y \in \mathcal{R}(Y)} P(X = x, Y = y) && \text{(definition of the joint PMF)} \end{aligned}$$

□

Example 0.19 (Summing a row of the joint PMF). In Example 0.17, summing the $X_1 = 1$ row:

$$\begin{aligned} P(X_1 = 1) &= P(X_1 = 1, X = 0) + P(X_1 = 1, X = 1) + P(X_1 = 1, X = 2) && \text{(marginal PMF from the joint PMF)} \\ &= 0 + \frac{1}{4} + \frac{1}{4} && \text{(read the table)} \\ &= \frac{1}{2} && \text{(add)} \end{aligned}$$

which matches $X_1 \sim \text{Ber}(1/2)$ from Example 0.6.

Definition 0.16 (Joint probability density function). For **jointly distributed** continuous random variables X and Y , a **joint probability density function** (joint density) of X and Y , denoted $f_{X,Y}(x,y)$ or $p(X = x, Y = y)$, is a function $f_{X,Y} \geq 0$ on $\mathbb{R}^2$ whose integral over any region $A \subseteq \mathbb{R}^2$ is the probability that the pair (X, Y) falls in A :

$$\Pr((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

As with [events](#), a fully rigorous version restricts A to a designated collection of regions; every region that arises in these notes is in it. Not every pair of continuous random variables has a joint density: if X is continuous, the pair (X, X) falls on the line $y = x$, a region of area 0, so every candidate $f_{X,Y}$ would give that line probability 0 instead of 1.

Example 0.20 (A joint density on a triangle). Let $f_{X,Y}(x, y) = 2$ for $0 \leq x \leq y \leq 1$, and 0 otherwise. The triangle $\{(x, y) : 0 \leq x \leq y \leq 1\}$ has area $1/2$, so $f_{X,Y}$ gives the whole plane probability $2 \cdot \frac{1}{2} = 1$, and it is the joint density of a pair (X, Y) with $X \leq Y$ always. For example, the probability that both are at most $1/2$ is 2 times the area of the smaller triangle $\{0 \leq x \leq y \leq 1/2\}$:

$$\Pr(X \leq \tfrac{1}{2}, Y \leq \tfrac{1}{2}) = 2 \cdot \tfrac{1}{8} = \tfrac{1}{4}$$

Theorem 0.6 (Marginal density from a joint density). *If X and Y have joint density $f_{X,Y}$, then X has the density:*

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

i Proof

Proof. For $a \leq b$, the event $\{a \leq X \leq b\}$ is the event that (X, Y) falls in the strip $[a, b] \times \mathbb{R}$, so:

$$\begin{aligned} \Pr(a \leq X \leq b) &= \Pr((X, Y) \in [a, b] \times \mathbb{R}) && \text{(same event)} \\ &= \iint_{[a, b] \times \mathbb{R}} f_{X,Y}(x, y) dx dy && \text{(definition of a joint density)} \\ &= \int_a^b \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right) dx && \text{(iterate the integral; Tonelli's theorem)} \\ &= \int_a^b f_X(x) dx && \text{(definition of } f_X) \end{aligned}$$

Tonelli's theorem allows the iterated integral because $f_{X,Y} \geq 0$ ([Fubini–Tonelli theorem](#); Billingsley (1995), Theorem 18.3). So f_X satisfies Definition [0.8](#). $\square$

Example 0.21 (Marginal densities on the triangle). For the joint density of Example 0.20 and $x \in [0, 1]$, $f_{X,Y}(x, y) = 2$ exactly when $x \leq y \leq 1$, so:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad (\text{marginal density from a joint density}) \\ &= \int_x^1 2 dy \quad (f_{X,Y}(x, y) = 2 \text{ for } x \leq y \leq 1, \text{ else } 0) \\ &= 2(1 - x) \quad (\text{integrate}) \end{aligned}$$

The same steps with the roles swapped give $f_Y(y) = \int_0^y 2 dx = 2y$ for $y \in [0, 1]$.

Definition 0.17 (Joint density-mass function). Let X be a discrete random variable and Y a continuous random variable, **jointly distributed**. A **joint density-mass function** of X and Y , denoted $p(X = x, Y = y)$, is a function $p(X = x, Y = y) \geq 0$ that is a probability mass in x and a probability density in y : for every x and every set $B \subseteq \mathbb{R}$,

$$\Pr(X = x, Y \in B) = \int_B p(X = x, Y = y) dy$$

Taking $B = \mathbb{R}$ gives $\int_{-\infty}^{\infty} p(X = x, Y = y) dy = P(X = x)$. When X is continuous and Y is discrete, the roles swap: $\Pr(X \in B, Y = y) = \int_B p(X = x, Y = y) dx$.

Example 0.22 (A coin flip and a waiting time). Let $p(X = 0, Y = y) = 1/2$ for $y \in [0, 1]$, $p(X = 1, Y = y) = 1/4$ for $y \in [0, 2]$, and 0 otherwise. Integrating over y gives $P(X = 0) = 1/2$ and $P(X = 1) = 1/2$, which add to 1. For example, the probability that $X = 1$ and $Y \leq 1$ is:

$$\Pr(X = 1, Y \leq 1) = \int_0^1 \frac{1}{4} dy = \frac{1}{4}$$

Survival, hazard, and cumulative hazard functions

Definition 0.18 (Survival function). The **survival function** (or **survivor function**) of a random variable T , denoted $S(t)$, is the probability that T exceeds t :

$$S(t) \stackrel{\text{def}}{=} \Pr(T > t)$$

The name comes from time-to-event analysis: if T is the time at which a participant dies, then $S(t)$ is the probability that the participant is still alive at time t . The definition itself applies

to any random variable.

Theorem 0.7 (Survival function and CDF). *For any random variable T with CDF $F(t)$:*

$$S(t) = 1 - F(t)$$

If T is continuous with density $f(t)$, then also:

$$S(t) = \int_{u=t}^{\infty} f(u) du$$

i Proof

Proof. The event $\{T > t\}$ is the complement of the event $\{T \leq t\}$, so the complement rule gives:

$$\begin{aligned} S(t) &\stackrel{\text{def}}{=} \Pr(T > t) && \text{(definition of the survival function)} \\ &= 1 - \Pr(T \leq t) && \text{(complement rule)} \\ &= 1 - F(t) && \text{(definition of the CDF)} \end{aligned}$$

For continuous T , the density integrates to 1 over the real line (Theorem 0.4), and F is the integral of the density up to t (Theorem 0.1), so:

$$\begin{aligned} 1 - F(t) &= \int_{u=-\infty}^{\infty} f(u) du - \int_{u=-\infty}^t f(u) du && \text{(total integral is 1; } F \text{ is the integral of } f) \\ &= \int_{u=t}^{\infty} f(u) du && \text{(split the first integral at } t \text{ and cancel)} \end{aligned}$$

□

Example 0.23 (Survival function of an exponential distribution). Let T be exponential with rate $\lambda > 0$ (Definition 0.10), so T has density $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$. For $t \geq 0$, by Theorem 0.7:

$$\begin{aligned} S(t) &= \int_{u=t}^{\infty} \lambda e^{-\lambda u} du && \text{(integral form of the survival function)} \\ &= [-e^{-\lambda u}]_{u=t}^{\infty} && \text{(antiderivative of } \lambda e^{-\lambda u}) \\ &= 0 - (-e^{-\lambda t}) && \text{(evaluate at the bounds; } e^{-\lambda u} \rightarrow 0 \text{ as } u \rightarrow \infty) \\ &= e^{-\lambda t} && \text{(simplify)} \end{aligned}$$

For $t < 0$, $S(t) = \Pr(T > t) = 1$, since $T \geq 0$. With $\lambda = 0.5$, for example, $S(2) = e^{-1} \approx$

0.368.

Definition 0.19 (Hazard function). The **hazard function** (also called the **hazard rate** or **hazard rate function**) of a continuous random variable T at a value t with $\Pr(T \geq t) > 0$, typically denoted $\lambda(t)$ or $h(t)$, is the limit of the **conditional probability** that T falls in an interval starting at t , given $T \geq t$, divided by the width of that interval, as that width shrinks to 0:

$$\lambda(t) \stackrel{\text{def}}{=} \lim_{\Delta \downarrow 0} \frac{\Pr(t \leq T < t + \Delta \mid T \geq t)}{\Delta}$$

Sources differ on the symbol: $h(t)$ appears in Dobson and Barnett (2018), Vittinghoff et al. (2012), Klein and Moeschberger (2003), and Kleinbaum and Klein (2012), while $\lambda(t)$ appears in Rothman et al. (2021) and Kalbfleisch and Prentice (2011).

If T is the time at which an event occurs, then $\lambda(t)$ is a rate, not a probability: for a small interval width $\Delta > 0$, $\lambda(t) \cdot \Delta$ is approximately the probability that the event occurs in $[t, t + \Delta)$, given that it has not occurred before t . A hazard can exceed 1, just as a density can. Many sources write the hazard as $\lambda(t) = p(T = t \mid T \geq t)$, reading it as the density of T at t , conditional on the event $T \geq t$; that notation abbreviates the limit in Definition 0.19. For a discrete T , the conditional probability $\Pr(T = t \mid T \geq t)$ itself is called the *discrete-time hazard*, and it is a probability.

The name “hazard” carries a connotation that the event is undesirable — death, relapse, equipment failure, and so on. When the event in question is neutral or desirable (recovery, conception, graduation, response to treatment), the same quantity $\lambda(t)$ is often called the **event incidence rate** instead. This terminology parallels the convention that conditional probabilities of undesirable events are called **risks**, while the same conditional probabilities for neutral or desirable events are simply called **probabilities**. The math is identical; only the name changes with the valence of the event.

Theorem 0.8 (Hazard equals density over survival). *If T is a continuous random variable with **density** $f(t)$ and **survival function** $S(t)$, then for every t with $S(t) > 0$ at which f is continuous:*

$$\lambda(t) = \frac{f(t)}{S(t)}$$

i Proof

Proof. The proof uses three facts. First, for $\Delta > 0$ the event $\{t \leq T < t + \Delta\}$ is a subset of the event $\{T \geq t\}$, so intersecting them leaves $\{t \leq T < t + \Delta\}$ (the **subset property**).

Second, $\Pr(T = t) = 0$ for a continuous T , so $\Pr(T \geq t) = \Pr(T > t) = S(t)$, which is positive. Third, because f is continuous at t , Theorem 0.3 gives $f(t)$ as the limit of $\Pr(t \leq T < t + \Delta)/\Delta$.

$$\begin{aligned}
\lambda(t) &\stackrel{\text{def}}{=} \lim_{\Delta \downarrow 0} \frac{\Pr(t \leq T < t + \Delta \mid T \geq t)}{\Delta} && \text{(definition of the hazard function)} \\
&= \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \cdot \frac{\Pr(\{t \leq T < t + \Delta\} \cap \{T \geq t\})}{\Pr(T \geq t)} && \text{(definition of conditional probability)} \\
&= \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \cdot \frac{\Pr(t \leq T < t + \Delta)}{\Pr(T \geq t)} && \text{(subset property)} \\
&= \frac{1}{\Pr(T \geq t)} \cdot \lim_{\Delta \downarrow 0} \frac{\Pr(t \leq T < t + \Delta)}{\Delta} && (\Pr(T \geq t) \text{ does not depend on } \Delta) \\
&= \frac{f(t)}{\Pr(T \geq t)} && \text{(density as a limit; } f \text{ is continuous at } t) \\
&= \frac{f(t)}{S(t)} && (\Pr(T = t) = 0 \text{ for continuous } T)
\end{aligned}$$

□

Example 0.24 (Hazard function of an exponential distribution). Continuing Example 0.23, for $t > 0$, where f is continuous:

$$\begin{aligned}
\lambda(t) &= \frac{f(t)}{S(t)} && \text{(hazard equals density over survival)} \\
&= \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} && \text{(substitute the exponential density and survival function)} \\
&= \lambda && \text{(cancel } e^{-\lambda t})
\end{aligned}$$

At $t = 0$, where f jumps from 0 to λ , Definition 0.19 gives the same value directly: $\Pr(0 \leq T < \Delta \mid T \geq 0)/\Delta = (1 - e^{-\lambda \Delta})/\Delta \rightarrow \lambda$ as $\Delta \downarrow 0$. So the exponential distribution has a constant hazard for $t \geq 0$, equal to its rate parameter; for $t < 0$, $f(t) = 0$, so $\lambda(t) = 0$.

Definition 0.20 (Cumulative hazard function). The **cumulative hazard function** of a continuous random variable T , often denoted $\Lambda(t)$ or $H(t)$, is the integral of its **hazard function** up to t :

$$\Lambda(t) \stackrel{\text{def}}{=} \int_{u=-\infty}^t \lambda(u) du$$

For a non-negative T , such as a time to event, $\lambda(u) = 0$ for $u < 0$, so the lower limit of the integral can be replaced by 0: $\Lambda(t) = \int_{u=0}^t \lambda(u) du$. That is the form most survival-analysis texts use.

Example 0.25 (Cumulative hazard function of an exponential distribution). Continuing Example 0.24, the hazard is $\lambda(u) = \lambda$ for $u \geq 0$ and 0 for $u < 0$, so for $t \geq 0$:

$$\begin{aligned}\Lambda(t) &= \int_{u=-\infty}^0 0 du + \int_{u=0}^t \lambda du && \text{(split the integral at 0)} \\ &= 0 + \lambda t && \text{(integrate each piece)} \\ &= \lambda t && \text{(simplify)}\end{aligned}$$

and $\Lambda(t) = 0$ for $t < 0$.

Corollary 0.1 (Survival function from the cumulative hazard). *If T is a continuous random variable whose density f is continuous at all but finitely many points, then for every t with $S(t) > 0$:*

$$\Lambda(t) = -\log\{S(t)\}$$

and therefore:

$$S(t) = \exp\{-\Lambda(t)\} \tag{1}$$

i Proof

Proof. Since $S(t) = 1 - F(t)$ (Theorem 0.7) and F has derivative f wherever f is continuous (Theorem 0.1), S has derivative $-f$ at those points. At each such u with $S(u) > 0$:

$$\begin{aligned}\frac{d}{du}(-\log\{S(u)\}) &= -\frac{1}{S(u)} \cdot \frac{d}{du} S(u) && \text{(chain rule)} \\ &= -\frac{1}{S(u)} \cdot (-f(u)) && (S' = -f) \\ &= \frac{f(u)}{S(u)} && \text{(simplify)} \\ &= \lambda(u) && \text{(hazard equals density over survival)}\end{aligned}$$

The function $-\log\{S(u)\}$ is continuous, because S is continuous for a continuous T , and it has derivative $\lambda(u)$ at all but finitely many points, so it is an antiderivative of λ for the fundamental theorem of calculus. As $u \rightarrow -\infty$, $S(u) \rightarrow 1$, so $-\log\{S(u)\} \rightarrow 0$. Therefore:

$$\begin{aligned}
\Lambda(t) &\stackrel{\text{def}}{=} \int_{u=-\infty}^t \lambda(u) du && \text{(definition of the cumulative hazard)} \\
&= [-\log\{S(u)\}]_{u=-\infty}^t && (-\log S \text{ is an antiderivative of } \lambda) \\
&= -\log\{S(t)\} - 0 && \text{(evaluate at the bounds)} \\
&= -\log\{S(t)\} && \text{(simplify)}
\end{aligned}$$

Exponentiating both sides of $-\Lambda(t) = \log\{S(t)\}$ gives Equation 1. $\square$

Example 0.26 (Recovering the exponential survival function from its cumulative hazard). Continuing Example 0.25, for $t \geq 0$:

$$\begin{aligned}
S(t) &= \exp\{-\Lambda(t)\} && \text{(survival function from the cumulative hazard)} \\
&= \exp\{-\lambda t\} && \text{(substitute } \Lambda(t) = \lambda t)
\end{aligned}$$

which matches the survival function computed directly in Example 0.23.

Table 1: Probability distribution functions of a continuous random variable T

Name	Symbols	Definition
Probability density function (PDF)	$f(t), p(t)$	$f \geq 0$ with $\int_a^b f(u) du = \Pr(a \leq T \leq b)$
Cumulative distribution function (CDF)	$F(t)$	$\Pr(T \leq t)$
Survival function	$S(t), \bar{F}(t)$	$\Pr(T > t)$
Hazard function	$\lambda(t), h(t)$	$\lim_{\Delta \downarrow 0} \Pr(t \leq T < t + \Delta \mid T \geq t) / \Delta$
Cumulative hazard function	$\Lambda(t), H(t)$	$\int_{u=-\infty}^t \lambda(u) du$
Log-hazard function	$\eta(t)$	$\log\{\lambda(t)\}$

For a continuous random variable $T \geq 0$ whose density is continuous at all but finitely many points, Theorem 0.7, Theorem 0.8, and Corollary 0.1 connect the functions in Table 1; each arrow in the following diagram converts one function into the next:

$$\begin{aligned}
f(t) &\xleftarrow[\frac{-S'(t)}{S(t)\lambda(t)}]{} S(t) \xleftarrow[\exp\{-\Lambda(t)\}]{} \Lambda(t) \xleftarrow[\int_{u=0}^t \lambda(u) du]{} \lambda(t) \xleftarrow[\exp\{\eta(t)\}]{} \eta(t) \\
f(t) &\xrightarrow[\int_{u=t}^{\infty} f(u) du]{} S(t) \xrightarrow[-\log\{S(t)\}]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t)
\end{aligned}$$

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